Airline delay analysis

Overview

Delay

Routes

AIRLINE

All

Origin Airport

All

Destination

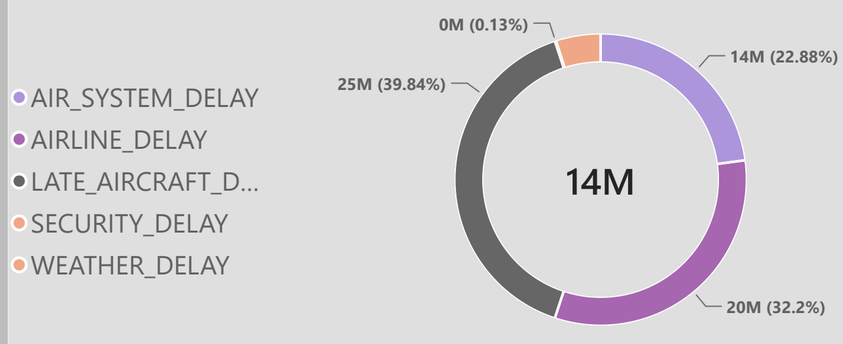
All

Origin Airport

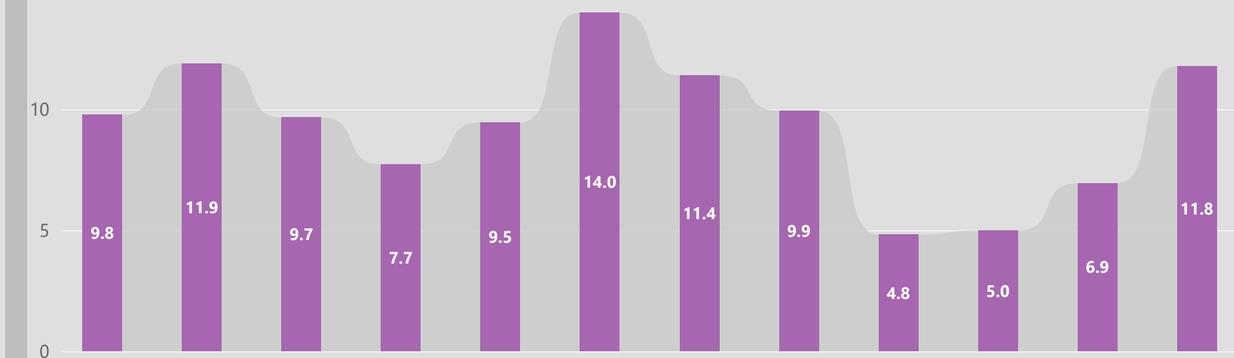
All

Total Flights	Avg Delay	Delay %	Cancellation %
6M	9.37	36.5%	1.5%

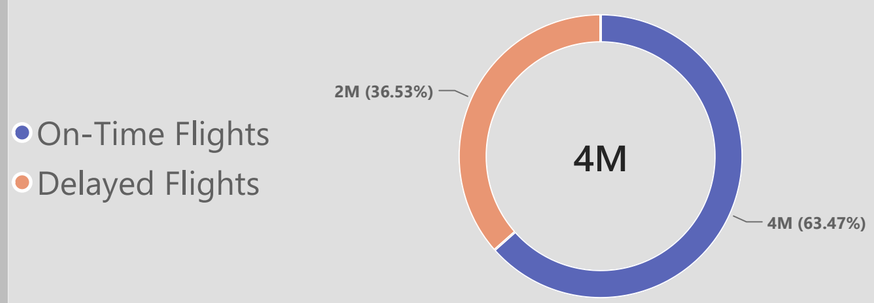
reasons of delays in minutes



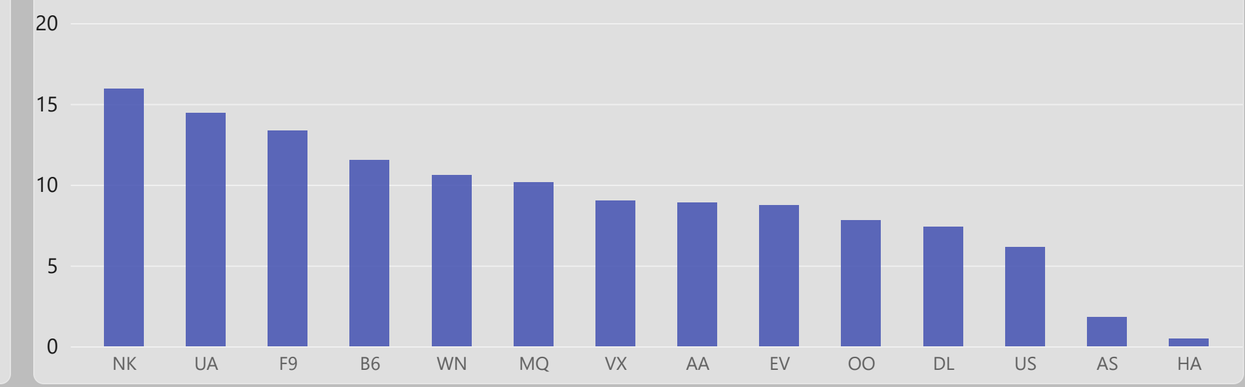
Avg Delay by MONTH



On-Time Flights and Delayed Flights



Average of DEPARTURE_DELAY by AIRLINE



36.5% of the 6M flights were delayed. Late Aircraft (39.8%) and Airline (32.2%) delays together drive 72% of total delay minutes. Delays peak in June (14.0 min avg) and December (11.8 min), while September–October are lowest (5 min).

Airline delay analysis

Overview

Delay

Routes

AIRLINE
All

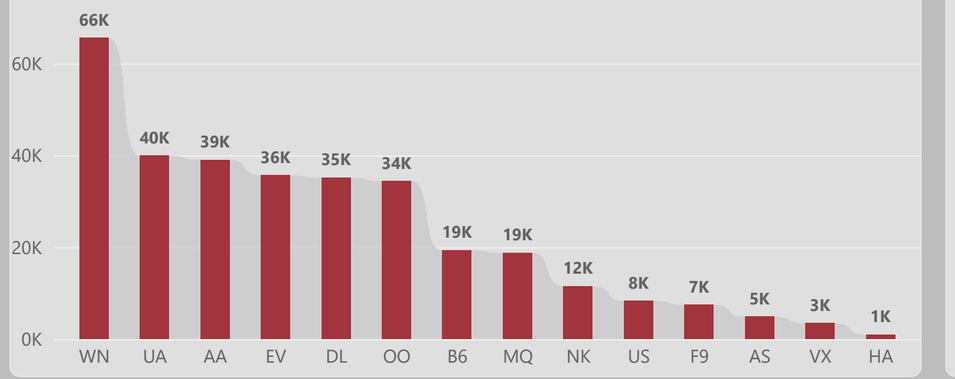
Origin Airport
All

Destination
All

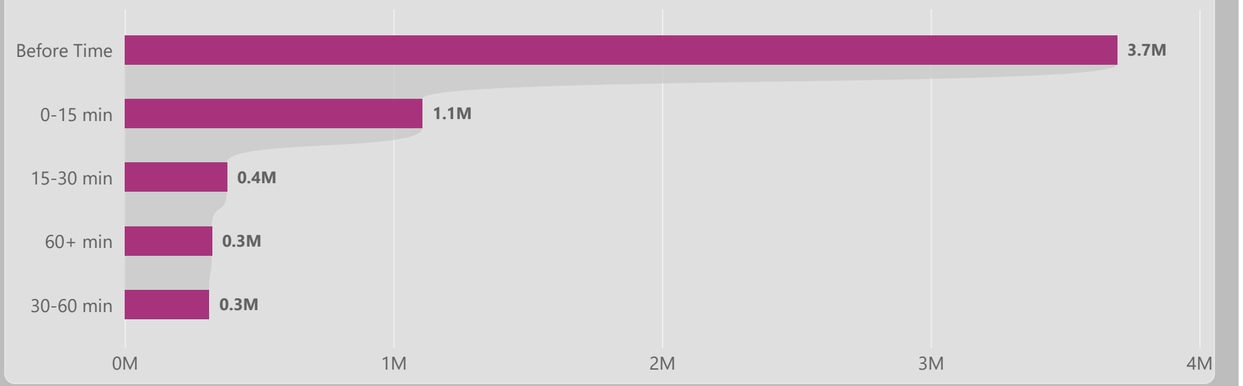
Day of week
All

Delayed Flights	On-Time Flights	Avg Delay	Severe Delays
2M	4M	9.37	324K

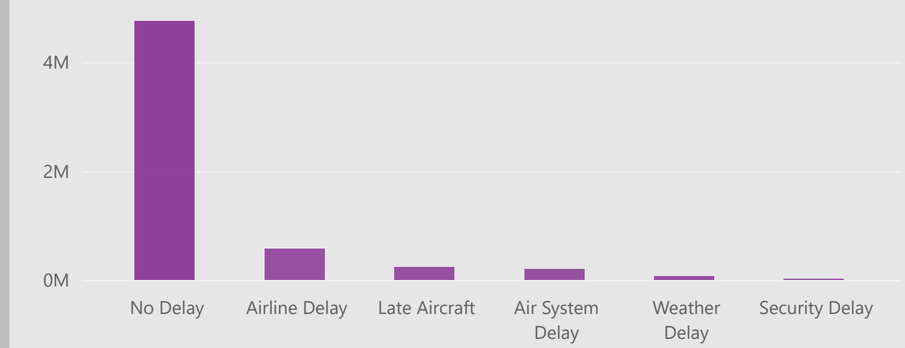
Airlines with Highest Severe Delays



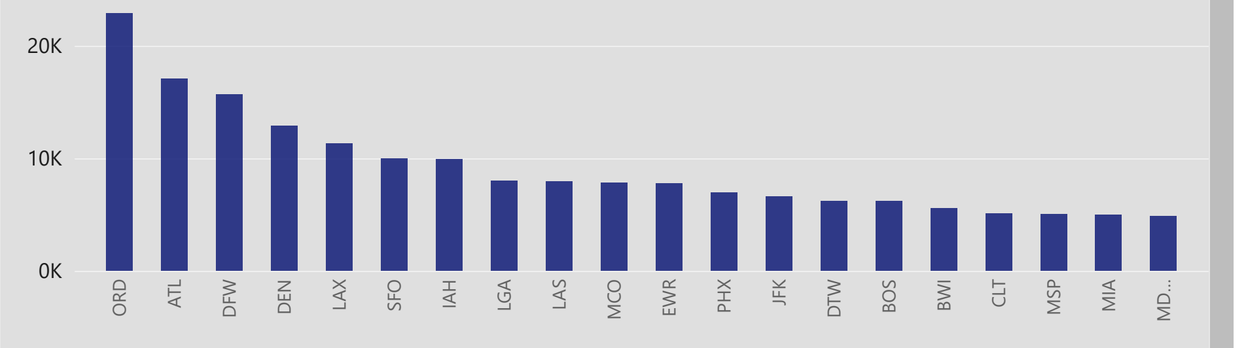
Flight Distribution by Delay Duration



What is causing flight delays



Which airports have most severe delays



Most delayed flights fall in the 0–15 min bucket, but 324K severe delays (60+ min) contribute disproportionately to total delay time. NK has the highest avg departure delay (16 min) while HA has the lowest

Airline delay analysis

Overview

Delay

Routes

AIRLINE
All

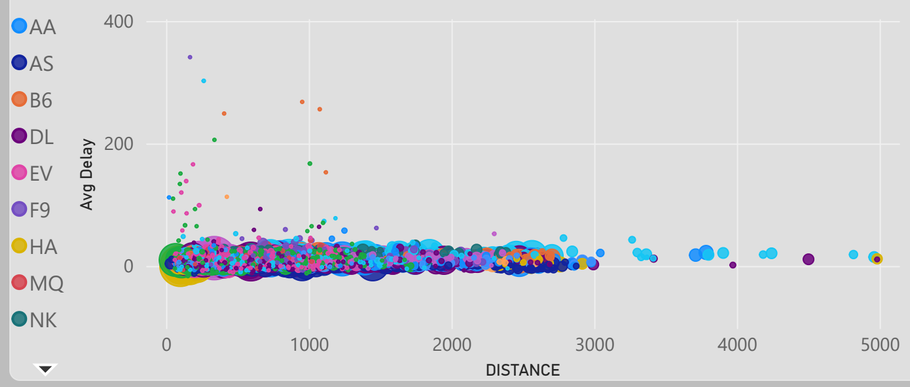
Origin Airport
All

Destination
All

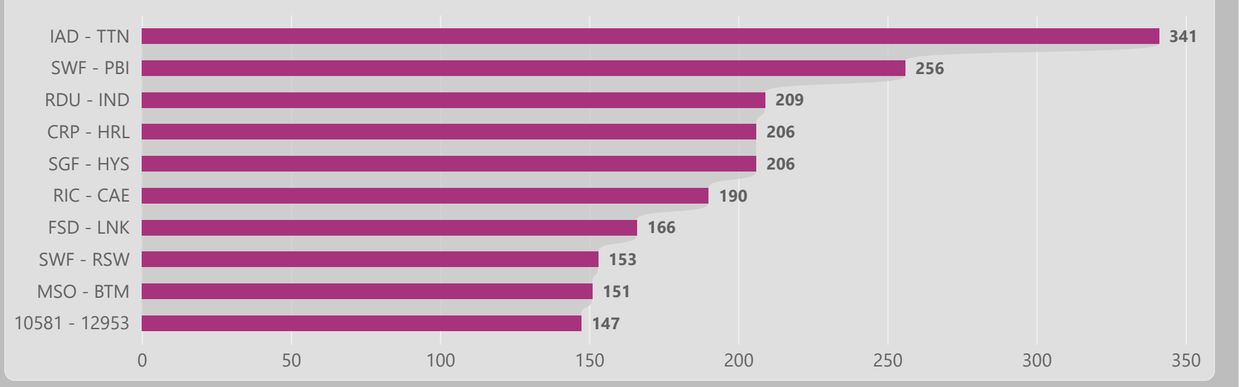
Day of week
All

Delayed Flights	On-Time Flights	Avg Delay	Severe Delays
2M	4M	9.37	324K

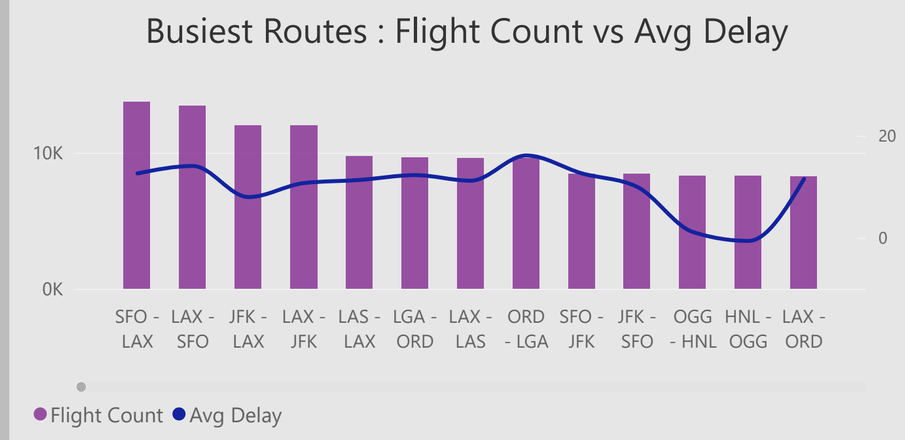
how much Distance Impact Flight Delays



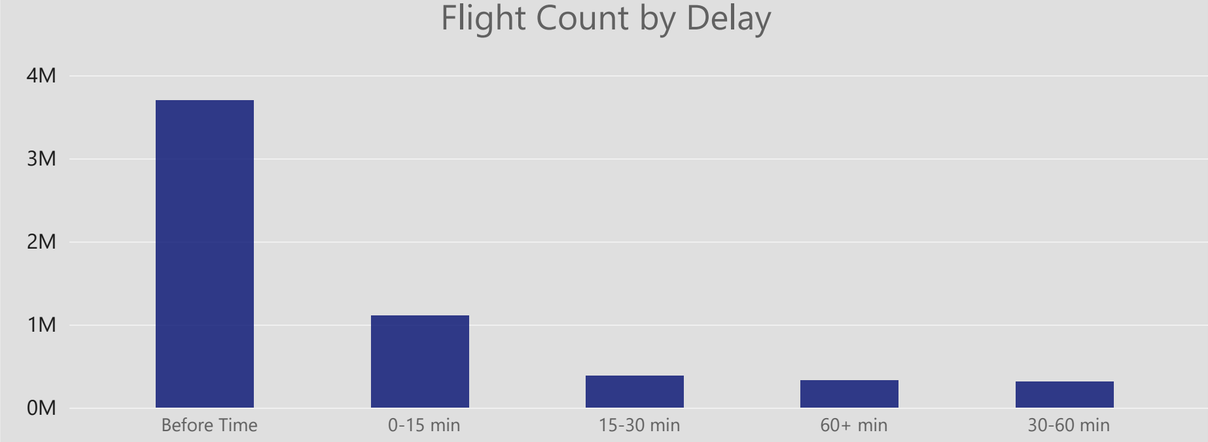
Top 10 Worst Routes



Busiest Routes : Flight Count vs Avg Delay



Flight Count by Delay



My analysis shows that most delays are short and operational in nature. Distance does not significantly impact delays, but high-traffic routes may face congestion issues. Airline and aircraft-related factors are the primary contributors, while severe delays, though fewer, have the highest business impact.